WORKED SOLUTION

STEP II 2002, Question 4

Pure • Inequalities from Integrals

STEP II 2002, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q4

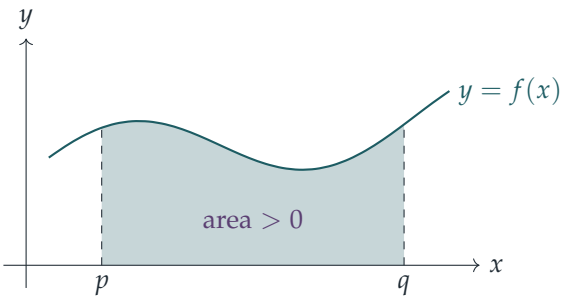
The key idea

A quadratic $as^2 - bs + c$ with $a > 0$ and $b^2 < 4ac$ has no real roots and a positive leading coefficient, so it is positive for *every* real s. Whatever we substitute for s (x, $\sin x$ or $1/x$), the resulting function is positive, so its integral over any interval is positive. Each part is then just a matter of choosing the substitution and the interval so that the integral produces exactly the numbers in the required inequality.

Solution

The opening sketch

The integral $\int_p^q f(x)\,\mathrm{d}x$ is the signed area between the curve $y = f(x)$ and the x-axis, from $x = p$ to $x = q$. Area above the axis counts as positive and area below counts as negative. If $f(x) > 0$ throughout $p < x < q$, the whole curve lies above the axis there, so the region has no negative part and has a genuine (non-zero) area. Hence the integral is strictly positive. (All the functions in this question are continuous, so the area is well defined.)



A fact used in every part

Let $g(s) = as^2 - bs + c$ with $a > 0$ and $b^2 < 4ac$. Completing the square,

$$g(s) = a\left(s - \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a}.$$

The first term is at least 0 because $a > 0$, and the second is strictly positive because $4ac - b^2 > 0$ and $a > 0$. So $g(s) > 0$ for all real s.

Part (i)

Take $f(x) = ax^2 - bx + c = g(x)$, which is positive for all x. We may integrate over any interval we like, and $[0, 1]$ gives the simplest numbers:

$$0 < \int_0^1 (ax^2 - bx + c)\,\mathrm{d}x = \frac{a}{3} - \frac{b}{2} + c.$$

Multiplying by 6 gives $2a - 3b + 6c > 0$, that is, $3b < 2a + 6c$, as required.

Part (ii)

Now $f(x) = a\sin^2 x - b\sin x + c = g(\sin x)$. Since g is positive at every real number, it is certainly positive at $\sin x$, so $f(x) > 0$ for all x.

? **Ask yourself** Which interval should we integrate over?

The target has π multiplying a and c but not b. Over $[0, \frac{\pi}{2}]$, $\int \sin x\,\mathrm{d}x = 1$ has no π, while $\int \sin^2 x\,\mathrm{d}x$ and $\int 1\,\mathrm{d}x$ do. Using $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$,

$$\int_0^{\pi/2} \sin^2 x\,\mathrm{d}x = \left[\frac{x}{2} - \frac{\sin 2x}{4}\right]_0^{\pi/2} = \frac{\pi}{4}, \qquad \int_0^{\pi/2} \sin x\,\mathrm{d}x = \Big[-\cos x\Big]_0^{\pi/2} = 1.$$

Hence

$$0 < \int_0^{\pi/2} f(x)\,\mathrm{d}x = \frac{a\pi}{4} - b + \frac{c\pi}{2}.$$

Multiplying by 4 gives $a\pi + 2c\pi - 4b > 0$, that is, $4b < (a + 2c)\pi$, as required.

Part (iii)

? **Ask yourself** What function has integrals giving $\ln(q/p)$, $\frac{1}{p} - \frac{1}{q}$ and $q - p$?

These are $\int_p^q \frac{1}{x}\,\mathrm{d}x$, $\int_p^q \frac{1}{x^2}\,\mathrm{d}x$ and $\int_p^q 1\,\mathrm{d}x$. So take

$$f(x) = \frac{a}{x^2} - \frac{b}{x} + c = g\left(\frac{1}{x}\right),$$

which is defined and continuous for $x > 0$, and positive there because g is positive everywhere. Since $q > p > 0$, the interval $[p, q]$ avoids $x = 0$, so

$$0 < \int_p^q \left(\frac{a}{x^2} - \frac{b}{x} + c \right) \mathrm{d}x = \left[-\frac{a}{x} - b \ln x + cx \right]_p^q$$
$$= a\left(\frac{1}{p} - \frac{1}{q} \right) - b \ln \frac{q}{p} + c(q - p).$$

Rearranging, $b \ln(q/p) < a\left(\frac{1}{p} - \frac{1}{q}\right) + c(q - p)$, as required. (The condition $p > 0$ also lets us write $\ln q - \ln p = \ln(q/p)$ with both logarithms defined.)

Checks

Take $a = b = c = 1$, so $b^2 = 1 < 4 = 4ac$. Then (i) says $3 < 8$, (ii) says $4 < 3\pi \approx 9.42$, and (iii) with $p = 1$, $q = 2$ says $\ln 2 \approx 0.693 < \frac{1}{2} + 1 = 1.5$. All hold. We also tested all three inequalities on 20 000 random triples (a, b, c) satisfying the conditions, including cases with b^2 just below $4ac$, and confirmed each integral against Simpson's rule. Note that the inequalities can fail without the conditions: $a = 1$, $b = 10$, $c = 1$ breaks (i), since $30 > 8$.

Common slips

- Forgetting to justify that $f > 0$: the conditions $a > 0$ and $b^2 < 4ac$ are exactly what makes the quadratic positive, and this has to be said.
- In (ii), taking $\int_0^{\pi/2} \sin^2 x \, \mathrm{d}x = \frac{\pi}{2}$ instead of $\frac{\pi}{4}$.
- In (iii), not mentioning that $p > 0$ keeps $x = 0$ (where f is undefined) out of the interval.

Beyond STEP

The principle "a positive continuous function has a positive integral" is one of the basic properties proved for the Riemann integral in a first analysis course. Together with linearity, it is the source of many classical inequalities, including the integral form of the Cauchy–Schwarz inequality, which comes from integrating the square $(f + tg)^2 \geq 0$ and using the same discriminant argument as here. Abbott's *Understanding Analysis* develops the integral carefully from this point of view.

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